

DATASCI 350 - SQL Essentials with DuckDB

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1 Introduction

SQL is no longer taught in the lectures of DATASCI 350. Module 06 now covers getting data from the web, and DATASCI 151 already introduces the basics of SQL. This self-study tutorial keeps the SQL material available for you. Read it if you want to add a widely used skill to your toolkit, or if you meet SQL in a job and want a reference written for this course.

The tutorial is self-contained. You will install one Python package, run every query yourself, and finish able to read and write the SQL you are most likely to see in a data science role. All the code here executes when the document is rendered, so every output you read below is real.

We use [DuckDB](#) as the engine. If you have met SQL before, it was probably through SQLite. DuckDB is a better fit for data science, and the reasons are worth a short section of their own.

2 Why learn SQL, and why DuckDB

SQL (Structured Query Language) is the common language of databases. You write a description of the data you want, and the database engine works out how to fetch it. The same language, with small dialect differences, drives SQLite on your laptop, PostgreSQL on a server, and BigQuery in the cloud. Learn it once and you can query almost any tabular data store you meet.

Two features make it worth your time. First, SQL is declarative: you say *what* you want, not *how* to compute it, so a short query can replace a long loop. Second, SQL is everywhere in data work. Analyst and data-scientist job adverts list it more often than any single Python library.

[DuckDB](#) is an SQL engine built for analytics. Three properties make it the right choice for this course:

- **Zero setup.** DuckDB is a single Python package. There is no server to start and no account to create. You run `pip install duckdb` and you have a working database.
- **It reads files directly.** DuckDB can query a CSV or Parquet file in place, with no import step. You point a query at a file and it returns rows.
- **It lives inside Python.** DuckDB runs in the same process as your Python code. It reads pandas and Polars DataFrames by name, and it returns results as DataFrames. SQL and Python stop being separate worlds.

You met DuckDB once already. Lecture 22 uses it, alongside Polars, as a tool for processing data at scale. This tutorial fills in the SQL underneath that lecture.

3 Setup

You install DuckDB the same way you install any Python package. The instructions below assume you have a working Python environment from earlier in the course.

Install the package first.

```
pip install duckdb
```

The tutorial also uses pandas and Polars in the final section. Install them too if they are missing.

```
pip install pandas polars pyarrow
```

Now verify the installation. Open Python. Run the two lines below. The version number appears.

```
import duckdb
print(duckdb.__version__)
```

1.5.5

You now have a working SQL engine. No server runs in the background. The engine is a library that your Python process loads.

3.1 Optional: the DuckDB command-line tool

You can also use DuckDB outside Python, in a terminal. The command-line tool is a separate download. It is optional for this tutorial, so skip this section if you only want the Python interface.

To install the command-line tool, visit the [DuckDB installation page](#). Follow the instructions for your operating system. To start it, type `duckdb` in a terminal. To leave it, type `.quit`.

The tool also has a browser interface. Start it with `duckdb -ui`. A local page opens where you can type SQL in a notebook cell and read the result as a table, as in the figure below. Everything in this tutorial works the same way, whether you type it in Python or in the command-line tool.

The screenshot shows the DuckDB browser interface. On the left, there's a sidebar with a search bar, a 'Notebooks' section showing an 'Untitled Notebook', and an 'Attached databases' section listing 'demo', 'main', and 'wdi'. The main area is titled 'Untitled Notebook' and contains a SQL query in a code cell. The query is: `SELECT country, round(value, 1) AS life_expectancy FROM wdi WHERE indicator = 'life_expectancy' AND year = 2019 ORDER BY value DESC LIMIT 8;`. Below the code cell, it says '8 rows returned in 7ms'. To the right of the code cell, there's a table with 2 columns: 'country' (VARCHAR) and 'life_expectancy' (DOUBLE). The table contains 8 rows of data, ordered by life expectancy in descending order.

	country	life_expectancy
1	Monaco	86.2
2	San Marino	85.3
3	Hong Kong SAR, China	85.2
4	Japan	84.4
5	Liechtenstein	84.2
6	Andorra	84.1
7	Macao SAR, China	83.9
8	Switzerland	83.9

Figure 1: The DuckDB browser interface, running one of this tutorial's queries against the World Bank data. Attached databases and their tables appear on the left.

4 Connecting to a database

A DuckDB database is either a file on disk or a temporary space in memory. You choose when you connect.

An **in-memory** database exists only while your program runs. It is fast and leaves nothing behind, which is ideal for learning and for one-off analysis. Connect with no argument.

```
con = duckdb.connect()
```

A **file-backed** database saves your tables to disk, so they survive after the program ends. Pass a filename to keep your work.

```
con = duckdb.connect("course.duckdb")
```

The only difference is the argument. Everything else in this tutorial works the same either way. We use an in-memory connection so the tutorial leaves no files behind.

You run a query with `con.sql()`. The result is a DuckDB *relation*. Add `.df()` to turn it into a pandas DataFrame, which prints as a familiar table.

```
con.sql("SELECT 'Hello, DuckDB' AS greeting, 2 + 2 AS sum").df()
```

```
      greeting  sum
0  Hello, DuckDB    4
```

We use `.df()` throughout so every result prints clearly. The last section shows the other ways to collect a result.

5 Databases and tables

A relational database stores data in **tables**. A table is a grid of rows and columns, like a single spreadsheet. Each column has a fixed **data type**, so a column of whole numbers cannot suddenly hold text. Each row is one record.

Two ideas connect tables to each other:

- A **primary key** is a column whose value is unique for every row. It identifies a record without ambiguity. A `driver_id` that is different for every driver is a primary key.
- A **foreign key** is a column in one table that points at the primary key of another. It is how a row in an orders table says which customer placed the order. Foreign keys are how relational databases avoid repeating the same information in many places.

DuckDB's core data types are few and predictable:

- **INTEGER** holds whole numbers.
- **DOUBLE** holds decimal numbers.
- **VARCHAR** holds text of any length.
- **BOOLEAN** holds true or false.
- **DATE** and **TIMESTAMP** hold calendar dates and times.
- **NULL** is not a type but a marker for a missing value. Any column can hold it.

5.1 Creating a table

You create a table with `CREATE TABLE`. You list each column and its type. Marking a column `PRIMARY KEY` tells DuckDB the values must be unique and present.

We build a small table of Formula 1 drivers, small enough to check by eye.

```
con.execute("""
CREATE TABLE drivers (
    driver_id    INTEGER PRIMARY KEY,
    driver_name  VARCHAR,
    team         VARCHAR,
    nationality  VARCHAR,
    victories    INTEGER
);
""")
```

Notice two conventions. SQL keywords are written in capitals, which separates them from your own column names. Each statement ends with a semicolon, which marks where one command stops and the next begins.

5.2 Inserting rows

You add rows with `INSERT INTO`. List the values for each row in the same order as the columns. Python's `None` becomes SQL's `NULL`, which is how we record that we do not know a driver's nationality or win count.

```
con.execute("""
INSERT INTO drivers VALUES
(1, 'Lewis Hamilton', 'Mercedes', 'British', 103),
(2, 'Max Verstappen', 'Red Bull Racing', 'Dutch', 55),
(3, 'Fernando Alonso', 'Aston Martin', NULL, NULL),
(4, 'Charles Leclerc', 'Ferrari', 'Monégasque', NULL),
(5, 'Valtteri Bottas', 'Mercedes', 'Finnish', 10),
(6, 'Sergio Pérez', 'Red Bull Racing', 'Mexican', 5),
(7, 'Lando Norris', 'McLaren', 'British', 4),
(8, 'Esteban Ocon', 'Alpine', 'French', 1);
""")
con.sql("SELECT * FROM drivers").df()
```

	driver_id	driver_name	team	nationality	victories
0	1	Lewis Hamilton	Mercedes	British	103
1	2	Max Verstappen	Red Bull Racing	Dutch	55
2	3	Fernando Alonso	Aston Martin	NaN	<NA>
3	4	Charles Leclerc	Ferrari	Monégasque	<NA>
4	5	Valtteri Bottas	Mercedes	Finnish	10
5	6	Sergio Pérez	Red Bull Racing	Mexican	5
6	7	Lando Norris	McLaren	British	4
7	8	Esteban Ocon	Alpine	French	1

The `SELECT * FROM drivers` at the end reads every column (*) and every row back. `SELECT` is the command you will use most. It extracts data without changing it.

5.3 Changing a table

Real tables change over time. A few commands cover most needs. Read the warning before you use `DROP TABLE`.

- Add a column: `ALTER TABLE drivers ADD COLUMN points INTEGER;`
- Remove a column: `ALTER TABLE drivers DROP COLUMN points;`
- Delete rows that match a condition: `DELETE FROM drivers WHERE victories = 0;`

Warning: `DROP TABLE` deletes the table and all its data at once. The action cannot be undone. Use it only when you are certain.

- Delete a whole table: `DROP TABLE drivers;`

We keep the drivers table, so we do not run those here.

6 Querying files directly

This is where DuckDB pulls ahead of a traditional database. You do not have to load a file into a table before you query it. You point a query straight at the file.

We use the course's World Bank data throughout the rest of the tutorial. It is the same dataset the final project pulls from the World Bank API, so the queries you practise here transfer directly.

The file `data/wdi_panel.parquet` ships with the tutorial. Copy it next to your own script first.

Copy the file `wdi_panel.parquet` into a folder named `data` beside your script. Confirm the path `data/wdi_panel.parquet` exists.

The data is in **long** format. Each row is one country, one indicator, and one year. Read five rows to see the shape.

```
con.sql("SELECT * FROM 'data/wdi_panel.parquet' LIMIT 5").df()
```

	country	iso3	indicator	year	value
0	Aruba	ABW	co2_per_capita	1990	3.203034
1	Aruba	ABW	co2_per_capita	1991	3.435717
2	Aruba	ABW	co2_per_capita	1992	3.634519
3	Aruba	ABW	co2_per_capita	1993	3.414535
4	Aruba	ABW	co2_per_capita	1994	3.683227

The quoted filename stands in for a table name. DuckDB opens the Parquet file, reads it, and treats it as a table for the length of the query.

Two commands describe a dataset before you query it. `DESCRIBE` lists the columns and their types.

```
con.sql("DESCRIBE SELECT * FROM 'data/wdi_panel.parquet'").df()
```

	column_name	column_type	null	key	default	extra
0	country	VARCHAR	YES	None	None	None
1	iso3	VARCHAR	YES	None	None	None
2	indicator	VARCHAR	YES	None	None	None
3	year	SMALLINT	YES	None	None	None
4	value	DOUBLE	YES	None	None	None

SUMMARIZE goes further. It reports counts, ranges, and the share of missing values per column, which is a fast way to learn a new dataset.

```
con.sql("""
SELECT column_name, min, max, count, null_percentage
FROM (SUMMARIZE SELECT country, indicator, year, value
      FROM 'data/wdi_panel.parquet')
""").df()
```

	column_name	min	max	count	null_percentage
0	country	Afghanistan	Zimbabwe	59024	0.00
1	indicator	co2_per_capita	urban_pop_pct	59024	0.00
2	year	1990	2023	59024	0.00
3	value	0.0	1438069596.0	59024	11.06

Typing the filename in every query is tedious. Register the file once as a **view**, a saved query that behaves like a table. From here on we write FROM wdi.

```
con.execute("""
CREATE VIEW wdi AS
SELECT * FROM 'data/wdi_panel.parquet';
""")
con.sql("SELECT COUNT(*) AS rows, COUNT(DISTINCT country) AS countries FROM wdi").df()
```

	rows	countries
0	59024	217

Which indicators does the dataset contain? A quick query answers it.

```
con.sql("SELECT DISTINCT indicator FROM wdi ORDER BY indicator").df()
```

	indicator
0	co2_per_capita
1	fertility_rate
2	gdp_per_capita

```

3     internet_users_pct
4     life_expectancy
5     population
6     primary_enrolment_net
7     urban_pop_pct

```

7 Core queries

Every SQL query is built from a few clauses in a fixed order. You select columns, filter rows, group them, and sort the result. This section covers each clause on both the small drivers table and the real World Bank data.

7.1 Selecting columns and filtering rows

SELECT chooses columns. WHERE keeps only the rows that meet a condition.

```

con.sql("""
SELECT driver_name, team, victories
FROM drivers
WHERE victories > 10
""").df()

```

	driver_name	team	victories
0	Lewis Hamilton	Mercedes	103
1	Max Verstappen	Red Bull Racing	55

Combine conditions with AND and OR. Parentheses make the logic explicit.

```

con.sql("""
SELECT driver_name, nationality, victories
FROM drivers
WHERE (nationality = 'British') AND (victories > 5)
""").df()

```

	driver_name	nationality	victories
0	Lewis Hamilton	British	103

On the World Bank data, the same clauses pull one indicator for one year. ORDER BY sorts the result, and DESC sorts from high to low. LIMIT keeps only the first rows.

```

con.sql("""
SELECT country, ROUND(value, 0) AS gdp_per_capita
FROM wdi
WHERE indicator = 'gdp_per_capita' AND year = 2020
ORDER BY value DESC
LIMIT 8
""").df()

```

	country	gdp_per_capita
0	Monaco	161263.0
1	Luxembourg	105274.0
2	Bermuda	98846.0
3	Isle of Man	86595.0
4	Switzerland	86294.0
5	Ireland	80928.0
6	Norway	78387.0
7	Cayman Islands	75200.0

7.2 Matching lists, ranges, and patterns

Three operators make WHERE conditions shorter and clearer.

IN checks whether a value is in a list. NOT IN excludes the list.

```
con.sql("""
SELECT driver_name, team
FROM drivers
WHERE team IN ('Ferrari', 'Mercedes')
""").df()
```

	driver_name	team
0	Lewis Hamilton	Mercedes
1	Charles Leclerc	Ferrari
2	Valtteri Bottas	Mercedes

BETWEEN checks a range, and it includes both endpoints.

```
con.sql("""
SELECT country, year, ROUND(value, 1) AS life_expectancy
FROM wdi
WHERE indicator = 'life_expectancy'
      AND country = 'Japan'
      AND year BETWEEN 2010 AND 2015
ORDER BY year
""").df()
```

	country	year	life_expectancy
0	Japan	2010	82.8
1	Japan	2011	82.6
2	Japan	2012	83.1
3	Japan	2013	83.3
4	Japan	2014	83.6
5	Japan	2015	83.8

LIKE matches text patterns. The % sign matches any run of characters, and _ matches exactly one character.

```
con.sql("""
SELECT DISTINCT country
FROM wdi
WHERE country LIKE 'United%'
""").df()
```

```

      country
0  United Arab Emirates
1      United States
2      United Kingdom
```

SQLite needed a `COLLATE NOCASE` trick to ignore case. DuckDB gives you `ILIKE`, which matches the same pattern while ignoring upper and lower case.

```
con.sql("SELECT driver_name FROM drivers WHERE driver_name ILIKE 'l%'").df()
```

```

      driver_name
0  Lewis Hamilton
1    Lando Norris
```

7.3 Aggregating

Aggregate functions reduce many rows to a single summary value. `COUNT`, `SUM`, `AVG`, `MIN`, and `MAX` are the common ones. `AS` names the result column, and `ROUND` controls decimal places.

```
con.sql("""
SELECT
    COUNT(*)           AS num_drivers,
    SUM(victories)     AS total_wins,
    ROUND(AVG(victories), 1) AS mean_wins,
    MAX(victories)     AS most_wins
FROM drivers
""").df()
```

```

num_drivers  total_wins  mean_wins  most_wins
0           8         178.0        29.7        103
```

`AVG` and the other aggregates skip `NULL` values automatically. The mean above divides by the number of non-missing wins, not by all eight drivers.

7.4 Grouping

`GROUP BY` splits rows into groups and computes an aggregate for each, returning one row per group. This is the workhorse of analysis. Here it gives the average value of each indicator for one year.

```
con.sql("""
SELECT indicator,
       COUNT(*)           AS n,
       ROUND(AVG(value), 2) AS mean_value
FROM wdi
WHERE year = 2020
GROUP BY indicator
ORDER BY indicator
""").df()
```

	indicator	n	mean_value
0	co2_per_capita	217	4.45
1	fertility_rate	217	2.49
2	gdp_per_capita	217	16149.94
3	internet_users_pct	217	64.11
4	life_expectancy	217	72.50
5	population	217	36084236.48
6	primary_enrolment_net	217	NaN
7	urban_pop_pct	217	61.82

To filter groups after aggregating, use HAVING. The difference from WHERE matters: WHERE filters rows before grouping, and HAVING filters groups after. You cannot put an aggregate in WHERE, so HAVING is where a condition on a group total belongs.

```
con.sql("""
SELECT team,
       COUNT(*)           AS drivers,
       SUM(victories)      AS team_wins
FROM drivers
GROUP BY team
HAVING SUM(victories) > 10
ORDER BY team_wins DESC
""").df()
```

	team	drivers	team_wins
0	Mercedes	2	113.0
1	Red Bull Racing	2	60.0

7.5 Try it yourself

Write queries against the wdi view for each task.

1. List the ten countries with the highest life expectancy in 2019.
2. Count how many countries have an internet_users_pct value recorded for the year 2000.
3. For the year 2015, report the average co2_per_capita grouped by whether the country's name starts with a letter before 'N'. Use a WHERE filter of your choice to keep it simple.

The solutions are in Section 12.

8 NULLs, CASE, and window functions

Real data is missing values, needs conditional labels, and often asks a question that GROUP BY cannot answer on its own. Three tools handle these cases.

8.1 Handling missing values

SQL marks a missing value as NULL. A NULL is not zero and not an empty string. It means “unknown”. Test for it with IS NULL or IS NOT NULL, never with =.

The World Bank data has real gaps. Count them for one indicator.

```
con.sql("""
SELECT
    COUNT(*)                AS total_rows,
    COUNT(value)            AS rows_with_value,
    SUM(CASE WHEN value IS NULL THEN 1 ELSE 0 END) AS missing
FROM wdi
WHERE indicator = 'primary_enrolment_net' AND year = 2020
""").df()
```

	total_rows	rows_with_value	missing
0	217	0	217.0

COALESCE returns the first value in its list that is not NULL. Use it to supply a default in place of a gap.

```
con.sql("""
SELECT driver_name, COALESCE(victories, 0) AS victories_filled
FROM drivers
ORDER BY driver_id
""").df()
```

	driver_name	victories_filled
0	Lewis Hamilton	103
1	Max Verstappen	55
2	Fernando Alonso	0
3	Charles Leclerc	0
4	Valtteri Bottas	10
5	Sergio Pérez	5
6	Lando Norris	4
7	Esteban Ocon	1

Here every missing win count becomes 0, so Fernando Alonso and Charles Leclerc now read as zero rather than blank.

8.2 Conditional logic with CASE

CASE is SQL's if-then-else. It reads each WHEN in order and returns the result of the first one that is true. The ELSE catches everything else. It is the natural way to turn a number into a label.

We classify countries by the World Bank's rough income bands, using GDP per capita in 2020.

```
con.sql("""
SELECT country, ROUND(value, 0) AS gdp_per_capita,
       CASE
         WHEN value >= 13846 THEN 'High income'
         WHEN value >= 4466  THEN 'Upper middle'
         WHEN value >= 1136  THEN 'Lower middle'
         WHEN value IS NULL  THEN 'No data'
         ELSE 'Low income'
       END AS income_band
FROM wdi
WHERE indicator = 'gdp_per_capita' AND year = 2020
      AND country IN ('Norway', 'Brazil', 'India', 'Burundi', 'China')
ORDER BY value DESC NULLS LAST
""").df()
```

	country	gdp_per_capita	income_band
0	Norway	78387.0	High income
1	China	10573.0	Upper middle
2	Brazil	8435.0	Upper middle
3	India	1807.0	Lower middle
4	Burundi	263.0	Low income

Order the WHEN clauses carefully. Because only the first true branch fires, a NULL check placed last would never run if an earlier branch matched. Put the most specific condition first.

8.3 Window functions

GROUP BY collapses each group to one row. A **window function** keeps every row and adds a value computed across a related set of rows. That is exactly what you need to rank rows, or to compare each row against its group's average, without losing the detail.

The OVER() clause defines the window. An empty OVER() uses the whole result. PARTITION BY splits it into groups. ORDER BY inside OVER() sets the order for ranking.

We rank countries by GDP per capita within one year, and show each country's gap from the global average that year.

```
con.sql("""
SELECT country,
       ROUND(value, 0) AS gdp_per_capita,
       RANK() OVER (ORDER BY value DESC) AS world_rank,
       ROUND(value - AVG(value) OVER (), 0) AS gap_from_mean
FROM wdi
```

```
WHERE indicator = 'gdp_per_capita' AND year = 2020
      AND value IS NOT NULL
ORDER BY world_rank
LIMIT 8
""").df()
```

	country	gdp_per_capita	world_rank	gap_from_mean
0	Monaco	161263.0	1	145113.0
1	Luxembourg	105274.0	2	89124.0
2	Bermuda	98846.0	3	82697.0
3	Isle of Man	86595.0	4	70445.0
4	Switzerland	86294.0	5	70144.0
5	Ireland	80928.0	6	64778.0
6	Norway	78387.0	7	62237.0
7	Cayman Islands	75200.0	8	59050.0

Compare the two approaches directly on the small drivers table. `GROUP BY team` would give one average win count per team and drop the driver names. A window function with `PARTITION BY team` attaches that same team average to every driver while keeping each driver's row.

```
con.sql("""
SELECT driver_name, team, victories,
       ROUND(AVG(victories) OVER (PARTITION BY team), 1) AS team_avg
FROM drivers
WHERE victories IS NOT NULL
ORDER BY team, victories DESC
""").df()
```

	driver_name	team	victories	team_avg
0	Esteban Ocon	Alpine	1	1.0
1	Lando Norris	McLaren	4	4.0
2	Lewis Hamilton	Mercedes	103	56.5
3	Valtteri Bottas	Mercedes	10	56.5
4	Max Verstappen	Red Bull Racing	55	30.0
5	Sergio Pérez	Red Bull Racing	5	30.0

8.4 Try it yourself

1. Use `RANK()` to rank countries by life expectancy in 2019, highest first, and show the top five.
2. Use `CASE` to label each country's 2020 internet_users_pct as 'High' (60 or more), 'Medium' (30 to 59), 'Low' (below 30), or 'No data'. Show ten rows.

The solutions are in Section 12.

9 Combining tables

Analysis usually needs data from more than one table. A **join** matches rows from two tables on a shared column. We add a small regions table by hand, then join it to the World Bank data.

First, build a subset of one indicator as a real table, so the joins have a clear left side.

```
con.execute("""
CREATE TABLE gdp2020 AS
SELECT country, iso3, ROUND(value, 0) AS gdp_per_capita
FROM wdi
WHERE indicator = 'gdp_per_capita' AND year = 2020
      AND iso3 IN ('USA', 'BRA', 'CHN', 'IND', 'DEU', 'FRA', 'ZAF', 'JPN');
""")
con.sql("SELECT * FROM gdp2020 ORDER BY iso3").df()
```

	country	iso3	gdp_per_capita
0	Brazil	BRA	8435.0
1	China	CHN	10573.0
2	Germany	DEU	42373.0
3	France	FRA	35709.0
4	India	IND	1807.0
5	Japan	JPN	35363.0
6	United States	USA	59534.0
7	South Africa	ZAF	5570.0

Now create a regions table. We give it a region for most of those countries, but leave Japan out on purpose, so the different joins produce visibly different results.

```
con.execute("""
CREATE TABLE regions (
    iso3    VARCHAR PRIMARY KEY,
    region  VARCHAR
);
INSERT INTO regions VALUES
('USA', 'North America'),
('BRA', 'Latin America'),
('CHN', 'East Asia'),
('IND', 'South Asia'),
('DEU', 'Europe'),
('FRA', 'Europe'),
('ZAF', 'Sub-Saharan Africa'),
('KOR', 'East Asia');
""")
con.sql("SELECT * FROM regions ORDER BY iso3").df()
```

	iso3	region
0	BRA	Latin America

1	CHN	East Asia
2	DEU	Europe
3	FRA	Europe
4	IND	South Asia
5	KOR	East Asia
6	USA	North America
7	ZAF	Sub-Saharan Africa

Note the asymmetry. `gdp2020` has Japan (JPN), which is missing from `regions`. `regions` has South Korea (KOR), which is missing from `gdp2020`. Each join treats these unmatched rows differently.

9.1 Inner join

An `INNER JOIN` keeps only rows that match in both tables. Japan and South Korea both drop out, because neither has a partner.

```
con.sql("""
SELECT g.country, r.region, g.gdp_per_capita
FROM gdp2020 AS g
INNER JOIN regions AS r ON g.iso3 = r.iso3
ORDER BY g.country
""").df()
```

	country	region	gdp_per_capita
0	Brazil	Latin America	8435.0
1	China	East Asia	10573.0
2	France	Europe	35709.0
3	Germany	Europe	42373.0
4	India	South Asia	1807.0
5	South Africa	Sub-Saharan Africa	5570.0
6	United States	North America	59534.0

The `g` and `r` after the table names are **aliases**, short nicknames that save typing and make the `ON` condition readable.

9.2 Left join

A `LEFT JOIN` keeps every row from the left table, matched or not. Where the right table has no match, its columns come back as `NULL`. Japan stays, with no region.

```
con.sql("""
SELECT g.country, r.region, g.gdp_per_capita
FROM gdp2020 AS g
LEFT JOIN regions AS r ON g.iso3 = r.iso3
ORDER BY g.country
""").df()
```

	country	region	gdp_per_capita
0	Brazil	Latin America	8435.0
1	China	East Asia	10573.0
2	France	Europe	35709.0
3	Germany	Europe	42373.0
4	India	South Asia	1807.0
5	Japan	NaN	35363.0
6	South Africa	Sub-Saharan Africa	5570.0
7	United States	North America	59534.0

The left join is the one you will reach for most, because it keeps the table you care about whole and adds detail where it exists.

9.3 Right and full joins

SQLite could not write these joins directly. It needed a swapped LEFT JOIN, and a UNION of two joins for the full case. DuckDB supports both natively, so the query says what it means.

A RIGHT JOIN keeps every row from the right table. South Korea now stays, with no GDP value.

```
con.sql("""
SELECT g.country, r.region, g.gdp_per_capita
FROM gdp2020 AS g
RIGHT JOIN regions AS r ON g.iso3 = r.iso3
ORDER BY r.region
""").df()
```

	country	region	gdp_per_capita
0	China	East Asia	10573.0
1	NaN	East Asia	NaN
2	Germany	Europe	42373.0
3	France	Europe	35709.0
4	Brazil	Latin America	8435.0
5	United States	North America	59534.0
6	India	South Asia	1807.0
7	South Africa	Sub-Saharan Africa	5570.0

A FULL OUTER JOIN keeps every row from both tables. Japan and South Korea both appear, each with NULL on the side that lacks a match.

```
con.sql("""
SELECT g.country, r.iso3, r.region, g.gdp_per_capita
FROM gdp2020 AS g
FULL OUTER JOIN regions AS r ON g.iso3 = r.iso3
ORDER BY r.region NULLS LAST
""").df()
```

	country	iso3	region	gdp_per_capita
0	China	CHN	East Asia	10573.0
1	NaN	KOR	East Asia	NaN
2	Germany	DEU	Europe	42373.0
3	France	FRA	Europe	35709.0
4	Brazil	BRA	Latin America	8435.0
5	United States	USA	North America	59534.0
6	India	IND	South Asia	1807.0
7	South Africa	ZAF	Sub-Saharan Africa	5570.0
8	Japan	NaN	NaN	35363.0

9.4 Cross and self joins

A **CROSS JOIN** pairs every row of one table with every row of the other. It has no **ON** condition. Use it to generate all combinations, such as every size with every colour.

```
con.execute("""
CREATE TABLE colours (colour VARCHAR);
CREATE TABLE sizes (size VARCHAR);
INSERT INTO colours VALUES ('Black'), ('Red');
INSERT INTO sizes VALUES ('S'), ('M');
""")
con.sql("""
SELECT colour, size, colour || ' - ' || size AS variant
FROM colours
CROSS JOIN sizes
ORDER BY colour, size
""").df()
```

	colour	size	variant
0	Black	M	Black - M
1	Black	S	Black - S
2	Red	M	Red - M
3	Red	S	Red - S

A **self join** joins a table to itself, which is how you follow a hierarchy stored in one table. A family table where `mother_id` points back to `person_id` is the classic example.

```
con.execute("""
CREATE TABLE family (
    person_id INTEGER PRIMARY KEY,
    name       VARCHAR,
    mother_id  INTEGER
);
INSERT INTO family VALUES
(1, 'Emma', NULL),
```

```

(2, 'Sarah', 1),
(3, 'Lisa', 1),
(4, 'Tom', 2),
(5, 'Alice', 2);
"""
)
con.sql("""
SELECT child.name AS child, mother.name AS mother
FROM family AS child
JOIN family AS mother ON child.mother_id = mother.person_id
ORDER BY mother.name
""").df()

```

```

      child mother
0  Sarah   Emma
1   Lisa   Emma
2    Tom  Sarah
3  Alice  Sarah

```

The two aliases `child` and `mother` are two views of the same table. The join matches each child's `mother_id` to a mother's `person_id`.

9.5 Combining rows: UNION, INTERSECT, EXCEPT

Joins add columns side by side. Set operators stack results on top of each other. Each query must return the same columns.

`UNION` stacks two results and removes duplicate rows. `UNION ALL` keeps duplicates and is faster.

```

con.sql("""
SELECT country, 'rich' AS tag FROM gdp2020 WHERE gdp_per_capita > 30000
UNION
SELECT country, 'poor' AS tag FROM gdp2020 WHERE gdp_per_capita < 10000
ORDER BY country
""").df()

```

```

      country  tag
0      Brazil poor
1      France rich
2     Germany rich
3        India poor
4         Japan rich
5  South Africa poor
6  United States rich

```

`INTERSECT` returns rows present in both results. `EXCEPT` returns rows in the first result that are absent from the second. Here they compare the country codes in the two tables.

```
con.sql("""
SELECT iso3 FROM gdp2020
INTERSECT
SELECT iso3 FROM regions
ORDER BY iso3
""").df()
```

```
iso3
0  BRA
1  CHN
2  DEU
3  FRA
4  IND
5  USA
6  ZAF
```

```
con.sql("""
SELECT iso3 FROM gdp2020
EXCEPT
SELECT iso3 FROM regions
""").df()
```

```
iso3
0  JPN
```

Only JPN is in gdp2020 but not in regions, which matches the asymmetry we built in.

9.6 Upsert: insert or update

Sometimes you want to insert a row, but update it instead if it already exists. SQL calls this an **upsert**, written as `INSERT ... ON CONFLICT`. It needs a `PRIMARY KEY` or `UNIQUE` column to detect the conflict. The excluded keyword refers to the row you tried to insert.

The regions table has `iso3` as its primary key. We reassign the United States to a new region. Because USA already exists, `DO UPDATE` runs instead of a failed insert.

```
con.execute("""
INSERT INTO regions VALUES ('USA', 'Americas')
ON CONFLICT (iso3) DO UPDATE SET region = excluded.region;
""")
con.sql("SELECT * FROM regions WHERE iso3 = 'USA'").df()
```

```
iso3    region
0  USA  Americas
```

DO NOTHING is the other option. It skips the insert quietly when the row already exists, which is useful when you load from several sources and want to keep the first version.

```
con.execute("""
INSERT INTO regions VALUES ('USA', 'This is ignored')
ON CONFLICT (iso3) DO NOTHING;
""")
con.sql("SELECT * FROM regions WHERE iso3 = 'USA'").df()
```

```
iso3    region
0  USA  Americas
```

The syntax here is identical to SQLite's. Upsert is one place where the two engines agree exactly.

9.7 Views

A **view** is a saved query that behaves like a table. It stores no data. It runs its query every time you read it. Views save you from rewriting a long query, and they give a stable name to a result even if the underlying tables change. You already made one: the wdi view.

Here is a view that joins gdp2020 to regions and averages GDP by region. Once created, anyone can query it without knowing the join.

```
con.execute("""
CREATE VIEW region_gdp AS
SELECT r.region,
       COUNT(*)           AS countries,
       ROUND(AVG(g.gdp_per_capita), 0) AS mean_gdp
FROM gdp2020 AS g
INNER JOIN regions AS r ON g.iso3 = r.iso3
GROUP BY r.region;
""")
con.sql("SELECT * FROM region_gdp ORDER BY mean_gdp DESC").df()
```

	region	countries	mean_gdp
0	Americas	1	59534.0
1	Europe	2	39041.0
2	East Asia	1	10573.0
3	Latin America	1	8435.0
4	Sub-Saharan Africa	1	5570.0
5	South Asia	1	1807.0

9.8 Try it yourself

1. Join gdp2020 and regions with an INNER JOIN, then report the single richest country in each region using MAX.
2. Use EXCEPT to list the country codes in regions that have no matching row in gdp2020.

The solutions are in Section 12.

10 DuckDB and DataFrames

DuckDB and pandas are not rivals. DuckDB reads a DataFrame by its Python variable name, and returns results as DataFrames, so you move between SQL and Python freely.

10.1 Querying a DataFrame with SQL

Put a DataFrame in a variable, then name it in a query as if it were a table. DuckDB finds it in your Python session.

```
import pandas as pd

drivers_df = con.sql("SELECT * FROM drivers").df()

con.sql("""
SELECT team, COUNT(*) AS n, ROUND(AVG(victories), 1) AS mean_wins
FROM drivers_df
GROUP BY team
ORDER BY n DESC
""").df()
```

	team	n	mean_wins
0	Mercedes	2	56.5
1	Red Bull Racing	2	30.0
2	McLaren	1	4.0
3	Aston Martin	1	NaN
4	Ferrari	1	NaN
5	Alpine	1	1.0

This is the feature that makes DuckDB comfortable inside a data science workflow. You clean data in pandas, run a fast SQL aggregation on it, and read the result straight back.

10.2 Returning results as pandas or Polars

A DuckDB relation converts to whichever frame you want. `.df()` gives pandas. `.pl()` gives Polars, the fast DataFrame library from Lecture 22. `.fetchall()` gives a plain list of tuples.

```
result = con.sql("""
SELECT country, ROUND(value, 1) AS life_expectancy
FROM wdi
WHERE indicator = 'life_expectancy' AND year = 2020
ORDER BY value DESC
LIMIT 3
""")

print("As tuples:", result.fetchall())
```

As tuples: [('Monaco', 86.1), ('Hong Kong SAR, China', 85.5), ('Japan', 84.6)]

The Polars conversion is one method call. The query below reshapes the long World Bank data into a wide table with pandas after the SQL step, which shows the two tools working together.

```
long_df = con.sql("""
SELECT country, indicator, value
FROM wdi
WHERE year = 2020
      AND country IN ('Brazil', 'Germany', 'Japan')
      AND indicator IN ('gdp_per_capita', 'life_expectancy', 'internet_users_pct')
""").df()

wide = long_df.pivot(index="country", columns="indicator", values="value").round(1)
wide
```

indicator	gdp_per_capita	internet_users_pct	life_expectancy
country			
Brazil	8435.0	81.3	74.5
Germany	42372.9	89.8	81.0
Japan	35363.3	90.2	84.6

10.3 When to use SQL, and when to use a DataFrame

Neither tool wins every time. A rough guide:

- Reach for **SQL** to join tables, to aggregate with GROUP BY, and to filter large files before loading them. These read clearly as one query, and DuckDB runs them quickly on data larger than memory.
- Reach for a **DataFrame** for row-by-row transformations, for plotting, for machine-learning input, and for anything that chains many small steps you want to inspect between.

Most real work mixes both. DuckDB makes the mixing cheap, because the data never leaves the process.

11 Going further

You now have the SQL you are most likely to need. To go deeper:

- The [DuckDB documentation](#) is clear and full of examples. The SQL reference lists every function.
- [DuckDB in Action](#) is a full book, free as a PDF from MotherDuck. It covers analytics, data pipelines, and the Python API in depth.
- The [Mode SQL tutorial](#) is a well-paced interactive course for general SQL practice.
- [SQLBolt](#) gives short interactive lessons in the browser, good for a quick refresher.

Lecture 22 returns to DuckDB as a tool for scaling and performance, alongside Polars. The SQL in this tutorial is the foundation that lecture builds on. When you meet a large Parquet or CSV file in the final project or in a job, you can now query it in place, join it, aggregate it, and hand the result to Python.

12 Solutions to exercises

The queries below answer the “Try it yourself” tasks. Try each task before you read its solution.

Core queries, task 1. Ten countries with the highest life expectancy in 2019.

```
con.sql("""
SELECT country, ROUND(value, 1) AS life_expectancy
FROM wdi
WHERE indicator = 'life_expectancy' AND year = 2019
ORDER BY value DESC
LIMIT 10
""").df()
```

	country	life_expectancy
0	Monaco	86.2
1	San Marino	85.3
2	Hong Kong SAR, China	85.2
3	Japan	84.4
4	Liechtenstein	84.2
5	Andorra	84.1
6	Macao SAR, China	83.9
7	Switzerland	83.9
8	Spain	83.8
9	Singapore	83.6

Core queries, task 2. Countries with an internet figure recorded in 2000.

```
con.sql("""
SELECT COUNT(*) AS countries_with_data
FROM wdi
WHERE indicator = 'internet_users.pct' AND year = 2000
AND value IS NOT NULL
""").df()
```

	countries_with_data
0	196

Core queries, task 3. Average CO2 per capita in 2015, split by an initial-letter condition.

```
con.sql("""
SELECT
    CASE WHEN country < 'N' THEN 'A to M' ELSE 'N to Z' END AS name_group,
    ROUND(AVG(value), 2) AS mean_co2
FROM wdi
WHERE indicator = 'co2_per_capita' AND year = 2015
GROUP BY name_group
ORDER BY name_group
""").df()
```

	name_group	mean_co2
0	A to M	4.09
1	N to Z	5.89

NULLs, CASE, windows, task 1. Top five countries by life expectancy in 2019 with RANK().

```
con.sql("""
SELECT country,
       ROUND(value, 1) AS life_expectancy,
       RANK() OVER (ORDER BY value DESC) AS rank
FROM wdi
WHERE indicator = 'life_expectancy' AND year = 2019 AND value IS NOT NULL
ORDER BY rank
LIMIT 5
""").df()
```

	country	life_expectancy	rank
0	Monaco	86.2	1
1	San Marino	85.3	2
2	Hong Kong SAR, China	85.2	3
3	Japan	84.4	4
4	Liechtenstein	84.2	5

NULLs, CASE, windows, task 2. Internet-use labels for 2020.

```
con.sql("""
SELECT country, ROUND(value, 1) AS internet_pct,
       CASE
         WHEN value IS NULL THEN 'No data'
         WHEN value >= 60 THEN 'High'
         WHEN value >= 30 THEN 'Medium'
         ELSE 'Low'
       END AS usage_band
FROM wdi
WHERE indicator = 'internet_users_pct' AND year = 2020
ORDER BY value DESC NULLS LAST
LIMIT 10
""").df()
```

	country	internet_pct	usage_band
0	United Arab Emirates	100.0	High
1	Bahrain	99.7	High
2	Qatar	99.7	High
3	Iceland	99.5	High
4	Kuwait	99.1	High
5	Monaco	98.6	High

6	Luxembourg	98.5	High
7	Saudi Arabia	97.9	High
8	Denmark	96.5	High
9	Korea, Rep.	96.5	High

Combining tables, task 1. Richest country in each region.

```
con.sql("""
SELECT r.region, MAX(g.gdp_per_capita) AS top_gdp
FROM gdp2020 AS g
INNER JOIN regions AS r ON g.iso3 = r.iso3
GROUP BY r.region
ORDER BY top_gdp DESC
""").df()
```

	region	top_gdp
0	Americas	59534.0
1	Europe	42373.0
2	East Asia	10573.0
3	Latin America	8435.0
4	Sub-Saharan Africa	5570.0
5	South Asia	1807.0

Combining tables, task 2. Region codes with no GDP row.

```
con.sql("""
SELECT iso3 FROM regions
EXCEPT
SELECT iso3 FROM gdp2020
ORDER BY iso3
""").df()
```

iso3
0 KOR

That is the end of the tutorial. You installed DuckDB, queried files in place, and worked through the core of SQL on real World Bank data. Keep this document as a reference, and open the [DuckDB documentation](#) when you need a function it did not cover.